Mini Project: Spam Email Detection using Logistic Regression

Title:

Spam Email Detection using Logistic Regression

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Objective:

To develop a simple machine learning model using **Logistic Regression** to classify emails as **spam** or **not spam** based on the text content.

Problem Statement:

Email spam is one of the most common forms of unsolicited communication. Filtering out spam helps improve user experience and security. This project aims to build a binary classifier that can automatically detect whether an email is spam or legitimate.

□ Data Source:

- **Kaggle** — SMS Spam Collection Dataset
  Link: https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset

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□ Libraries Used:

- pandas, numpy

- sklearn (Logistic Regression, TF-IDF, Metrics)
- nltk (stopwords, lemmatizer)
- matplotlib, seaborn
- Flask (for deployment)

 Preprocessing:

- Converted text to lowercase
- Removed punctuation, numbers, and stopwords
- Tokenized and lemmatized text
- Applied TF-IDF vectorization with bigrams

 Model:

- **Classifier:** Logistic Regression
- **Vectorization:** TF-IDF (max_features=3000, bigrams)
- **Train-Test Split:** 70% - 30%

 Evaluation Metrics:

Metric	Score
Accuracy	95.93%
Precision	98.14%
Recall	70.98%
F1 Score	82.38%
ROC AUC	85.38%

 Feature Importance:

Top indicative spam words: *free*, *win*, *claim*, *click*
Top indicative ham words: *ok*, *meet*, *call*

 Test Examples:

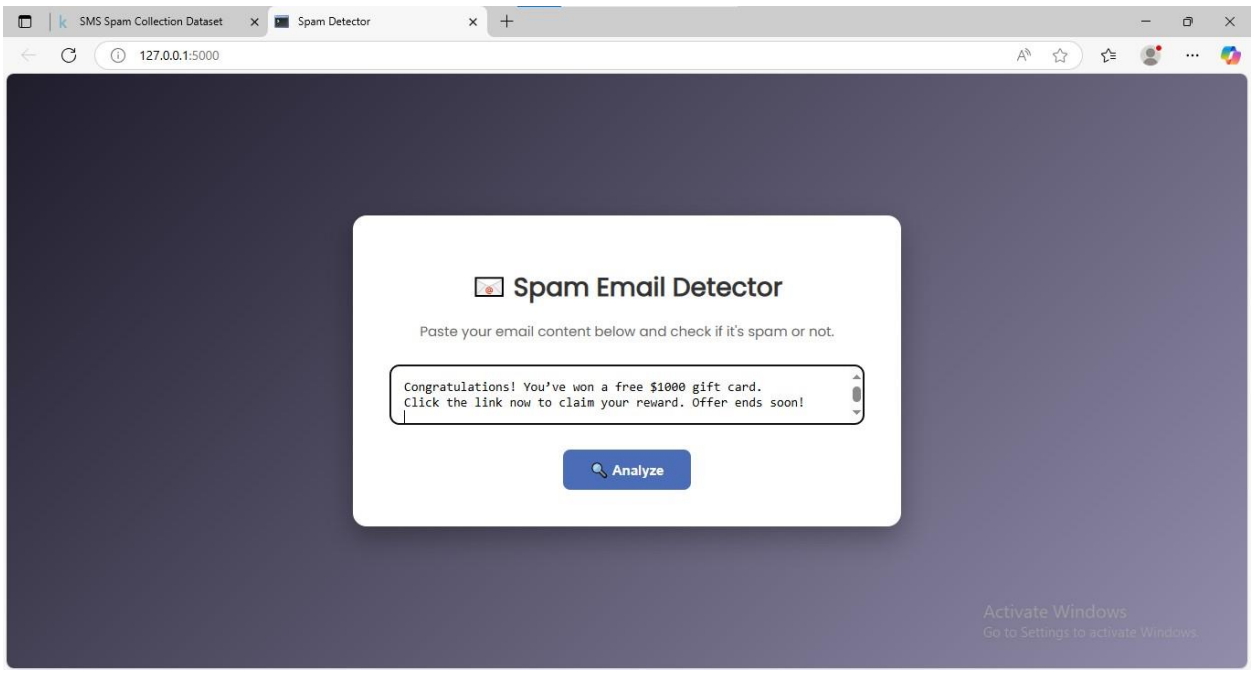
1. "Click the link to win a free iPhone!" → **SPAM**
2. "See you at the office by 3 PM" → **NOT SPAM**

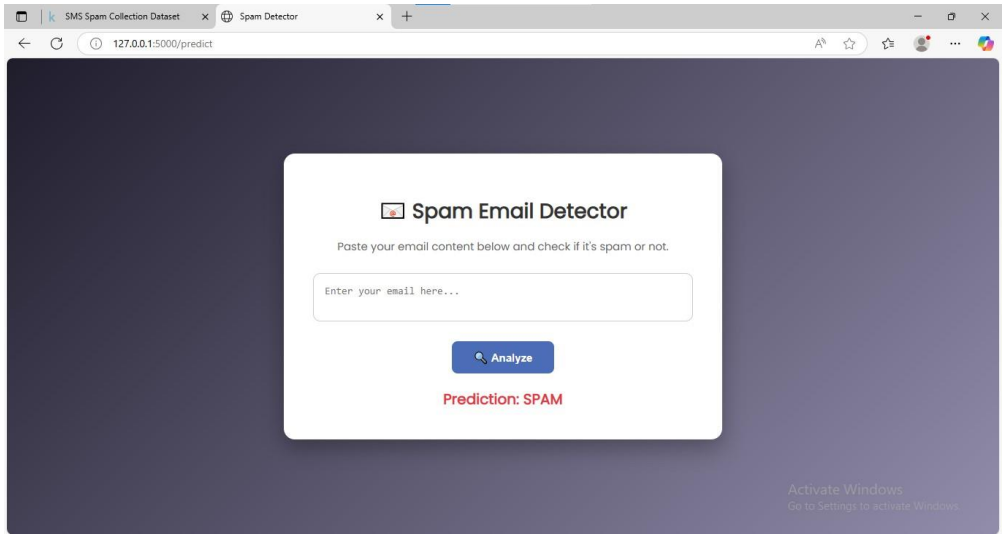
- Deployment:

- Built a **Flask web app**
- UI built using **HTML & CSS**
- Users can input email text and get real-time prediction

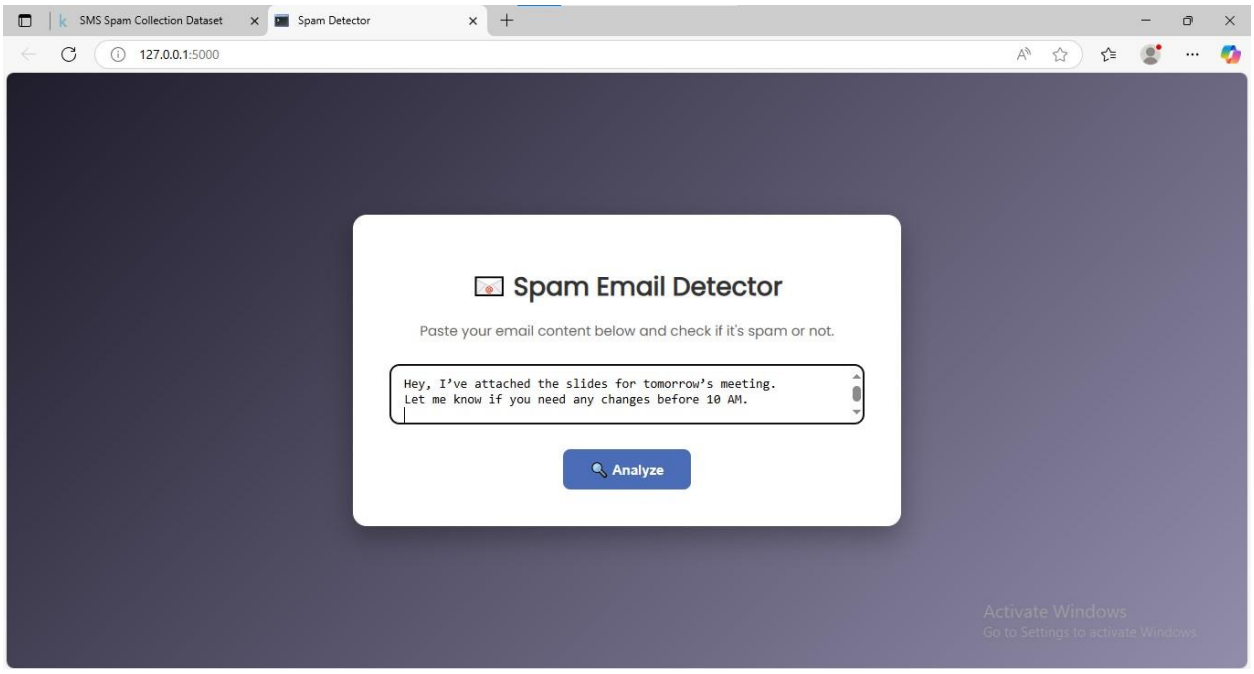
□ Sample UI Screenshots:

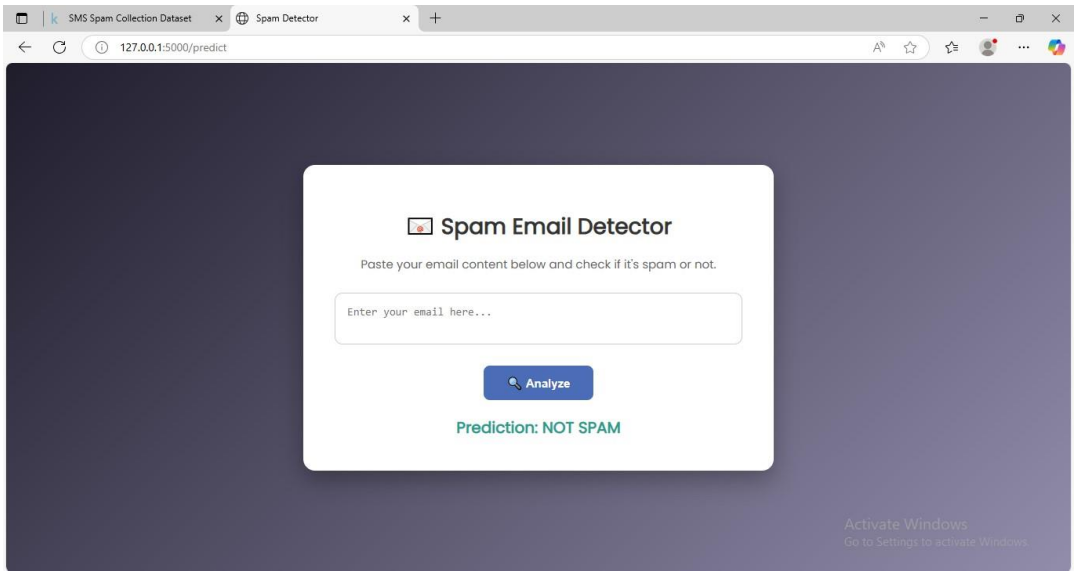
Example 1: SPAM—





Example 2: NOT SPAM—





❖ Conclusion:

The logistic regression model performed well with strong accuracy and precision. While recall can be improved, the system is highly interpretable and suitable for light deployment. With further enhancements, this can evolve into a more robust email filtering tool.